

Examiner
only

3. Explain how cosmic microwave background radiation (CMBR) **and** cosmological red shift provide evidence for the origin of the Universe. [6 QER]

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6. The following equation shows a fusion reaction that happens in stars.

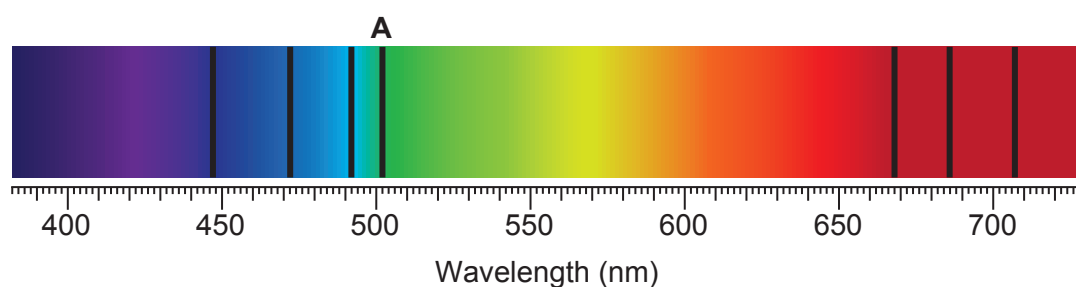


(a) State the **two** conditions needed for a fusion reaction to take place in a star. [2]

1.

2.

(b) An incomplete absorption spectrum from the Sun is shown below.



(i) An absorption line at wavelength $590 \times 10^{-9} \text{ m}$ is missing from the spectrum. **Complete the diagram** above by adding the missing absorption line. [1]

(ii) Describe **and** explain how the position of the absorption line labelled **A** would be different for a star in a more distant galaxy. [3]

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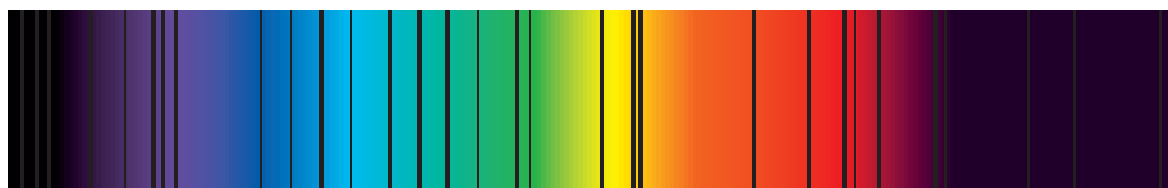
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5. The diagram shows the absorption spectrum of light from the Sun.



- (a) Explain how the dark lines are formed **and** how the spectrum can be used to identify the elements present in the Sun. [3]

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- (b) Many scientists believe that the Universe began with the Big Bang, 15.5 billion years ago.
State **two** pieces of evidence for the Big Bang **and** explain how they support this theory. [4]

Evidence 1:

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Evidence 2:

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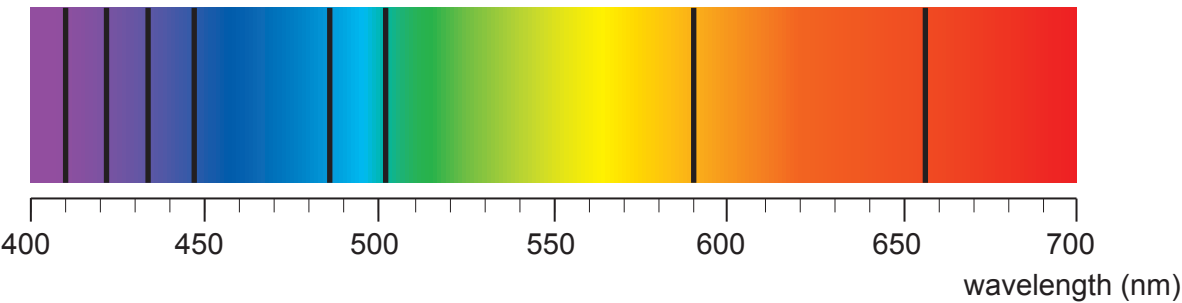
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4. The diagram shows a spectrum from a star.
The spectrum is crossed by dark lines.



The table gives information about the wavelengths of lines to be found in a spectrum if the element is present in the atmosphere of the star.

Element	Wavelength (nm)
helium	447, 502
iron	431, 467, 496, 527
hydrogen	410, 434, 486, 656
sodium	590

- Explain how the dark lines arise on the spectrum.
- Explain which elements are present in the star's atmosphere.

[6 QER]

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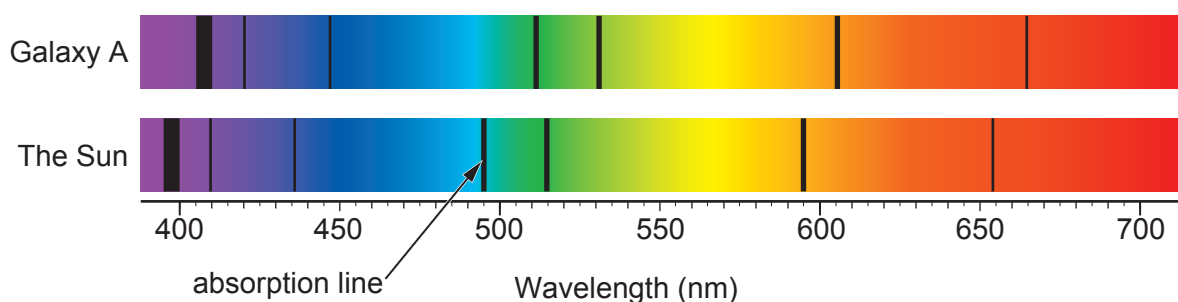
7. (a) (i) State the two pieces of evidence that support the Big Bang model of the origin of the Universe. [2]

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- (ii) The diagram below shows the absorption spectrum of a galaxy, A, compared to part of the Sun's absorption spectrum.



One absorption line in the spectrum from the Sun is labelled.
The wavelength of this line is different in galaxy A.
Explain why the wavelength of the line is different in galaxy A.

[2]

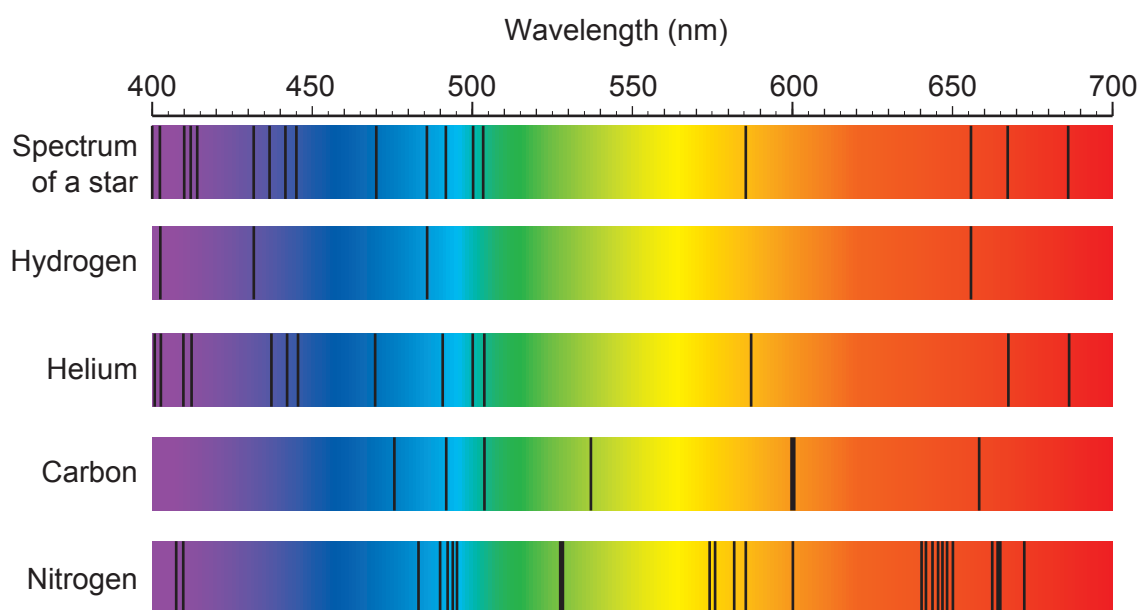
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- (b) The diagram below shows the absorption spectrum of a star and the absorption spectra of some different elements.



Matthew suggests that the star contains hydrogen and carbon.
Explain whether you agree.

[2]

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